

Progress-Aware Predictive Safety Filtering for Hierarchical Recurrent Reinforcement Learning in Multi-Vessel ASV Navigation

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Abstract—This paper presents an algorithmic hierarchical autonomy architecture for variable-speed path following and collision avoidance of an underactuated autonomous surface vessel (ASV) in mixed static and dynamic maritime environments. The approach combines separate recurrent Proximal Policy Optimization (PPO) policies for steering and speed, a recurrent joint residual policy that coordinates corrections to both commands, and a receding-horizon predictive safety filter that evaluates static and moving hazards before execution. The filter searches constant and structured emergency steering-speed sequences, maintains short-term obstacle memory, and includes a soft starboard preference in risky head-on encounters. To address conservative stop-heavy behavior observed in a representative post-training mixed-hazard scenario, a progress term is added only to the ranking of already-admissible actions, leaving the safety criterion unchanged. In simulation, the final seven-family validation suite achieved 87/90 successes (96.67%, Wilson 95% CI: 90.65–98.86%) with no stalls. The single-/multi-vessel, variable-speed, and COLREG-style encounter families achieved 55/55 successful episodes. On a separate fixed 14-episode controlled subset, the full method achieved 14/14 collision-free completions, compared with 9/14 and five collisions for nominal recurrent PPO and 10/14 with two collisions plus two cross-track-limit terminations for a classical LOS+CPA/TCPA baseline. Additional post-training generalization tests on newly defined path and obstacle layouts also completed successfully; six representative scenarios are reported in detail. Mean learned-policy inference was 2.73 ms, and a separate CPU profiling experiment identified predictive-search latency as the principal target for real-time optimization. The autonomy software was developed against the supplied target-ASV characteristics and transferred for subsequent hardware-in-the-loop and sea-trial validation; the experimental results reported here are the simulation-validation stage of that engineering workflow.

I. INTRODUCTION

Autonomous surface vessels must combine route following, speed regulation, and collision avoidance while operating under changing encounter geometry, uncertain target motion, and environmental disturbance. These objectives are tightly coupled: steering away from a target changes cross-track error and relative bearing, while slowing or accelerating can increase or reduce the available collision-avoidance maneuver set. International Regulations for Preventing Collisions at Sea (COLREGs) further influence maneuver choice, particularly in head-on, crossing, and overtaking encounters [1].

Deep reinforcement learning (DRL) has shown promise for marine path following and collision avoidance, including end-to-end PPO guidance, risk-based COLREG formulations, and multi-ship decision policies [9]–[11], [14], [15]. These studies establish that learned policies can jointly reason about path

following and encounter geometry, but their collision-avoidance behavior is primarily embedded in the learned policy or reward. In contrast, predictive safety filtering provides an explicit model-based action screen [7], [8]; this separation can improve constraint handling, but it can also introduce conservatism or chattering when the safety layer repeatedly overrides a nominal controller. Residual reinforcement learning offers a complementary mechanism for refining nominal behavior without discarding it [6].

This work studies an integrated alternative. Separate recurrent speed and steering policies provide nominal behavior; a joint recurrent residual policy coordinates bounded steering-speed corrections; and a predictive safety filter performs online candidate rollouts before selecting the command that is actually applied. The filter is intentionally outside residual-policy training, enabling the learned policies and explicit safety logic to remain modular. A progress-aware cost is then introduced only after candidate admissibility has been established, reducing unnecessary stopping without weakening the underlying clearance test.

The principal methodological contribution is a hierarchical variable-speed ASV autonomy architecture in which specialized recurrent steering and speed policies are coordinated by bounded learned joint residual correction and followed by receding-horizon joint action screening. A progress-aware objective ranks only safety-admissible actions, preserving the explicit clearance test while reducing conservative low-speed behavior. The contributions are: (1) the hierarchical recurrent steering-speed and residual architecture; (2) a predictive safety layer for static and multi-vessel encounters with obstacle memory and structured emergency maneuvers; (3) progress-aware ranking applied only after safety admissibility; and (4) a seven-family stress-test evaluation, controlled component ablation, a classical non-learning comparison, and post-training generalization tests. The reported experiments validate the autonomy algorithm in simulation using the target platform’s supplied operating characteristics; the transferred controller is intended for subsequent HIL integration and controlled water-trial validation.

II. RELATED WORK

Prior ASV/USV DRL work provides important reference points but solves a different architectural problem. Meyer *et al.* [10] used PPO for integrated path following and COLREG-aware collision avoidance, while Heiberg *et al.* [11]

incorporated collision-risk theory into a DRL guidance policy. Zhao and Roh [9] and subsequent multi-ship studies [14]–[16] similarly encode encounter geometry or COLREG-related preferences in learned decision making. These approaches demonstrate the viability of learned maritime guidance, but safety is chiefly represented through policy training, reward shaping, or encounter logic. Our objective is different: the learned steering-speed proposal is subjected to an independent finite-horizon clearance test before execution, so the online safety layer can reject a learned action without retraining the policies.

Predictive safety filters formalize this nominal-controller/safety-layer separation [7]. Recent safe-RL work also shows that safety-filter/controller interaction can degrade performance through intervention-induced conservatism and motivates explicitly accounting for this interaction [8]. Residual RL, meanwhile, learns bounded corrections around a nominal controller [6]. We combine these ideas in a maritime variable-speed setting: two specialized recurrent nominal policies are coordinated by a recurrent joint residual policy, after which a receding-horizon filter screens joint steering-speed candidates. The progress term is deliberately placed *after* admissibility testing, separating “is this candidate predicted safe?” from “which safe candidate best preserves motion?” The contribution is therefore the safety-oriented integration and ordering of these components rather than a modification of PPO itself. Because published ASV studies use different dynamics, encounter sets, and success definitions, they provide methodological context rather than directly transferable numerical baselines. We instead include a conventional LOS+CPA/TCPA non-learning baseline under the same fixed evaluation subset used for component ablation.

III. PROBLEM FORMULATION

A. Reduced-order ASV model

The simulated ASV is represented by planar position (x, y) , heading ψ , yaw rate r , and surge speed U . Yaw dynamics follow a first-order Nomoto model [5],

$$\dot{r} = \frac{K\delta - r}{T}, \quad \dot{\psi} = r, \quad (1)$$

with $K = 1.0$, $T = 2.0$ s, and $|\delta| \leq 35^\circ$. The simulation step is $\Delta t = 0.1$ s. Vessel length is 1.4 m and the collision radius is 0.7 m. The speed action $a_U \in [-1, 1]$ maps to

$$U_{\text{cmd}} = \frac{a_U + 1}{2} U_{\text{max}}, \quad U_{\text{max}} = 1.5 \text{ m/s}. \quad (2)$$

The surge-speed interface is limited to 0–1.5 m/s, with approximately 0.25 m/s² acceleration and 0.60 m/s² deceleration, reflecting the operating limits supplied for the target ASV. These limits are applied consistently in the plant update and predictive rollouts; detailed propeller/thrust dynamics are outside the controller model used in this study.

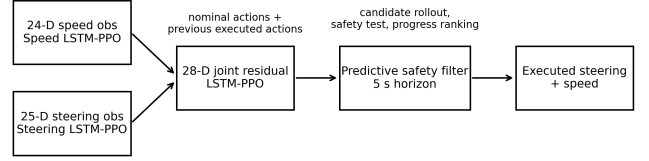


Fig. 1. Hierarchical controller. Separate recurrent speed and steering policies produce nominal actions. A recurrent joint residual policy forms a corrected proposal, which is screened by the predictive safety filter before execution.

Collision safety is measured using surface-to-surface clearance. For target j with center distance d_j and collision radius R_j ,

$$c_j = d_j - (R_s + R_j), \quad (3)$$

where R_s is the ASV collision radius. A negative clearance indicates geometric overlap.

B. CPA/TCPA and encounter representation

For relative position $p_r = p_o - p_s$ and velocity $v_r = v_o - v_s$, time to closest point of approach (TCPA) and center-distance at CPA are

$$t_{\text{CPA}} = -\frac{p_r^T v_r}{\|v_r\|^2}, \quad (4)$$

$$d_{\text{CPA}} = \|p_r + t_{\text{CPA}} v_r\|, \quad (5)$$

with surface DCPA computed by subtracting the vessel radii. Practical encounter classification uses relative bearing and heading difference. Head-on cases use approximately $|\beta| \leq 15^\circ$ and heading difference $\Delta\psi \geq 150^\circ$; crossing give-way and stand-on sectors lie approximately on starboard and port respectively; overtaking uses $|\beta| > 112.5^\circ$ with own speed exceeding target speed by more than about 0.05 m/s. These thresholds are control features, not a formal legal decision engine.

IV. HIERARCHICAL RECURRENT CONTROLLER

Fig. 1 summarizes the architecture. All learned components use recurrent PPO with LSTM state [2]–[4]. The nominal speed and steering policies are frozen before training the joint residual policy.

A. Nominal recurrent policies

The final speed policy uses a 24-D observation. Its first 12 features summarize own-vessel motion, path tracking, and LiDAR risk: normalized current and commanded speed, absolute cross-track error, $\sin(e_\psi)$, $\cos(e_\psi)$, normalized yaw rate, segment progress, minimum LiDAR range, forward/port/starboard minimum range, and normalized turn demand. Features 13–24 encode normalized TCPA, predicted surface clearance at CPA, current surface clearance, relative speed, sine/cosine of relative bearing, a collision-risk indicator, and one-hot encounter type (none, head-on, crossing give-way, crossing stand-on, overtaking). LiDAR quantities are normalized using a

TABLE I
FINAL RECURRENT PPO CONFIGURATIONS

	Speed	Steering	Residual
Observation / action	24 / 1	25 / 1	28 / 2
LSTM hidden	128	128	256
LSTM layers	1	1	1
MLP after LSTM	128,128	128,64	64,64
n_{steps}	512	1024	1024
Batch size	128	128	128
Epochs	10	10	10
γ	0.99	0.995	0.995
GAE λ	0.95	0.95	0.95
Entropy coeff.	0.004	0.001	0.003
Final LR	5×10^{-5}	5×10^{-5}	5×10^{-5}
Selected steps	390,544	680,000	225,000

30 m range; relative speed is normalized against approximately 3 m/s.

The final steering policy is speed aware and uses a 25-D observation: the original 24-D steering/navigation observation (six navigation states plus 18 radial LiDAR sectors) with normalized current speed inserted after the first six navigation features. The original steering features comprise normalized cross-track error, sine/cosine heading error, normalized yaw rate, previous normalized rudder, normalized path-segment progress, and 18 LiDAR sectors.

B. Joint residual correction

The residual policy receives

$$o_t^{\text{res}} = [o_t^{\text{speed}}, a_{\delta,t}^n, a_{U,t}^n, a_{\delta,t-1}^e, a_{U,t-1}^e] \in \mathbb{R}^{28}, \quad (6)$$

where superscripts n and e denote nominal and previously executed commands. Its 2-D output $[a_{\delta,r}, a_{U,r}] \in [-1, 1]^2$ produces bounded proposals

$$a_{\delta,c} = \text{clip}(a_{\delta,n} + 0.40a_{\delta,r}, -1, 1), \quad (7)$$

$$a_{U,c} = \text{clip}(a_{U,n} + 0.50a_{U,r}, -1, 1). \quad (8)$$

These are candidate commands; the safety filter may replace them. The residual layer is intended to coordinate two independently specialized nominal policies: it can jointly trade steering and speed corrections before the explicit safety screen, rather than forcing the safety filter to reconcile two unrelated nominal commands by itself.

C. Residual reward

Residual training excludes the predictive safety filter. The frozen nominal policies provide actions, the residual policy modifies them, and the environment adds shaping to the base navigation reward:

$$R = R_{\text{base}} + 0.01R_{\text{shape}}. \quad (9)$$

The shaping terms penalize underspeed, prolonged stopping, unnecessary steering magnitude/saturation/rate/sign flips, and residual magnitude, while rewarding speed recovery. Verified default weights are 10.0 (underspeed), 8.0 (prolonged stop), 2.0 (recovery), 3.0 (steering magnitude), 10.0 (saturation), 2.5 (steering rate), 6.0 (steering flip), and 0.35 (residual regularization). In particular,

$$P_{\text{res}} = 0.35(a_{\delta,r}^2 + a_{U,r}^2). \quad (10)$$

The final selected residual checkpoint was obtained at 225k steps from a configured 300k-step refinement run.

D. Online control cycle

At runtime the controller executes the following sequence at each step: (i) update navigation, LiDAR, and tracked-target features; (ii) query the recurrent speed and steering policies; (iii) append nominal and previous executed actions to form the 28-D residual observation; (iv) compute bounded residual corrections; (v) roll out the corrected proposal and candidate steering-speed pairs over the prediction horizon; (vi) reject candidates that violate the clearance criterion; (vii) rank safe candidates using deviation, smoothness, and progress terms; (viii) invoke structured emergency search when the constant-action set contains no admissible candidate; and (ix) execute only the first selected action before repeating the process. This separation is important because the residual policy never directly bypasses the safety layer.

V. TRAINING PROCEDURE

Training was performed in stages so that each higher-level component refined a fixed lower-level controller rather than changing the entire stack simultaneously. Intermediate stages used Stable-Baselines3 evaluation callbacks with periodic held-out evaluation, and the best evaluation checkpoint was retained for the next refinement stage. We do not infer that checkpoint selection optimized collision rate specifically; it followed the evaluator reward used by the training scripts.

The speed controller was developed through three sequential stages: 200k steps of initial variable-speed training (seed 12001, learning rate 3×10^{-4} , entropy coefficient 0.01), a 100k-step policy refinement (seed 61001, learning rate 10^{-4} , entropy 0.003), and a 150k-step deadlock-recovery refinement (seed 72001, learning rate 5×10^{-5} , entropy 0.004). The selected final speed checkpoint reports 390,544 accumulated timesteps because training continued from the preceding stages. During speed-policy development, the target speed was risk dependent; higher collision risk reduced the target-speed ratio, while low-risk conditions favored speeds near U_{max} . The final policy is learned and is not simply a hand-coded risk-to-speed map.

Speed-aware steering was trained in a 600k-step Stage 1 (seed 82001, learning rate 3×10^{-4} , entropy 0.005) and a 250k-step Stage-2 smooth refinement (seed 83001, learning rate 5×10^{-5} , entropy 0.001). The retained checkpoint is at 680k accumulated timesteps. The joint residual controller was then trained with both nominal policies frozen, using seed 86001, a configured 300k-step budget, learning rate 5×10^{-5} , and entropy coefficient 0.003. The final engineering stack retained the 225k residual checkpoint because it was the selected validation checkpoint, rather than automatically using the terminal 300k model.

The base navigation reward contains step cost, cross-track and heading-error penalties, progress reward, yaw-rate and rudder regularization, obstacle-related penalty/shaping, and waypoint/success terms. Because that reward evolved across earlier development stages, we avoid presenting a single

TABLE II
TRAINING LINEAGE OF FINAL LEARNED CONTROLLERS

Stage	Budget	Seed	Eval cadence
Speed initial training	200k	12001	10k / 20 eps
Speed refine	100k	61001	10k / 30 eps
Speed deadlock refine	150k	72001	10k / 30 eps
Steering Stage 1	600k	82001	20k / 30 eps
Steering Stage 2	250k	83001	10k / 30 eps
Residual joint	300k	86001	10k / 30 eps

collapsed equation that would imply one immutable reward for all three final policies. In contrast, the residual shaping described above is directly verified in the final residual environment.

VI. PREDICTIVE SAFETY FILTER

At each 0.1 s control step, the corrected proposal is evaluated over a $H = 5.0$ s prediction horizon. The nominal candidate grid contains 17 steering and 9 speed values; the exact proposal can also be inserted, so the unique candidate count can exceed the Cartesian-grid count. A candidate is admissible when the predicted static/dynamic clearance satisfies the required computational margin $M = 0.20$ m over the rollout. The filter applies only the first command of the selected sequence and replans on the next step.

Among safe candidates, the final ranking objective is

$$J = (a_\delta - a_{\delta,n})^2 + 0.35 \left(\frac{U - U_n}{U_{\max}} \right)^2 + 0.20(a_\delta - a_{\delta,prev})^2 + 5.0 \left(1 - \frac{U}{U_{\max}} \right)^2. \quad (11)$$

The last term is the progress-aware addition. Importantly, it is evaluated only after the candidate passes the safety test; it therefore changes preference among admissible actions rather than reducing the clearance requirement. Safety and progress are thus lexicographically separated: predicted clearance determines admissibility first, and progress only orders the surviving set.

A 0.15 m release margin provides hysteresis. The filter also maintains dynamic-target memory for 30 steps and static-target memory for 120 steps, with a 2.5 m association distance and bounded uncertainty-radius growth. Selective robust prediction can be triggered by abnormal tracked motion (approximately 0.75 m position residual, 0.35 m/s speed change, or 12° heading change) and maintained for 15 steps.

If no constant candidate is safe, structured emergency sequences are searched. Steering magnitudes include $\{\pm 1, \pm 0.75, \pm 0.5, \pm 0.25\}$ in normalized units, speed ratios include $\{0, 0.25, 0.5, 0.75, 1\}$, and first-stage durations include 0.50, 0.75, 1.50, and 2.50 s. Patterns include turn-and-continue, turn-then-straighten, S-turn/counter-turn, and low-speed turn followed by acceleration. When no fully safe sequence exists, fallback ranking emphasizes short-horizon survival margin, full-horizon margin, progress, and objective deviation. A soft port-turn penalty is applied in risky head-on cases, thereby favoring starboard maneuvers consistent with the direction

implied by COLREG Rule 14 without claiming formal Rules 13–17 compliance.

VII. EXPERIMENTAL SETUP

The final validation suite comprises seven named stress families using the same frozen learned policies and safety filter. For compact reproducibility we retain the identifiers R1–R7, but each identifier denotes a specific test family: R1, one moving vessel; R2, two moving vessels; R3, three-vessel stress; R4, tracking robustness with three to five vessels; R5, environmental-disturbance sweeps from calm through mismatch conditions; R6, variable-speed encounters; and R7, COLREG-style multi-vessel geometries including head-on/crossing combinations and mixed random encounters. Target-vessel speeds are primarily 1–2.5 m/s.

Disturbance is generated with an Ornstein-Uhlenbeck process ($\theta = 0.18$) and stochastic gusts lasting approximately 1.0–3.9 s. Representative wind/current mean–standard-deviation pairs progress from (0, 0.015)/(0, 0.010) in calm conditions to (0.18, 0.080)/(0.15, 0.060) in mismatch conditions; gust probability increases from 0 to 0.03 per step.

An episode counts as success only if it terminates without collision, stall, or truncation and satisfies the environment success flag; a fallback check requires the final waypoint and less than 8 m goal distance. Time-limit truncation is not counted as success. The common environment also terminates an episode for excessive route departure when $|\text{CTE}| > 20$ m while collision risk is below 0.2.

Post-training generalization was also probed on newly defined path and obstacle layouts. Six representative scenarios are documented in detail: a clean circular path, a circle with four static hazards, a synthetic shoreline with five moving vessels, a three-island archipelago, a mixed three-rock/two-vessel case, and a circular route with mixed static and moving hazards including a partially on-path obstacle.

VIII. RESULTS

A. Stress-test regression

Table III and Fig. 2 summarize the final R1–R7 validation. The controller completed 87/90 episodes successfully (96.67%; Wilson 95% CI 90.65–98.86%) with no stalls. R1–R3, R6, and R7 achieved 55/55 successful episodes across the principal single-/multi-vessel, variable-speed, and COLREG-style encounter families. The three failures occurred in the robustness-focused families: R4 tracking robustness achieved 14/15 and R5 environmental disturbance achieved 18/20.

One R4 collision (seed 160403) had minimum true surface clearance -0.0431 m. At the terminal event the safety filter already predicted a negative margin (approximately -0.3354 m), while tracked target position and velocity agreed with ground truth. This supports a no-safe-candidate/maneuver-reachability interpretation rather than an optimistic terminal tracking estimate. R5 retained an 18/20 disturbance result; a later diagnostic with a longer 6 s horizon recovered one previously failing seed, showing that some disturbance failures

TABLE III
FINAL R1–R7 VALIDATION REGRESSION

Family	Success	Rate
R1: one moving vessel	10/10	100%
R2: two moving vessels	10/10	100%
R3: three-vessel stress	10/10	100%
R4: tracking robustness	14/15	93.3%
R5: disturbance sweep	18/20	90.0%
R6: variable speed	5/5	100%
R7: COLREG-style encounters	20/20	100%
Overall	87/90	96.67%

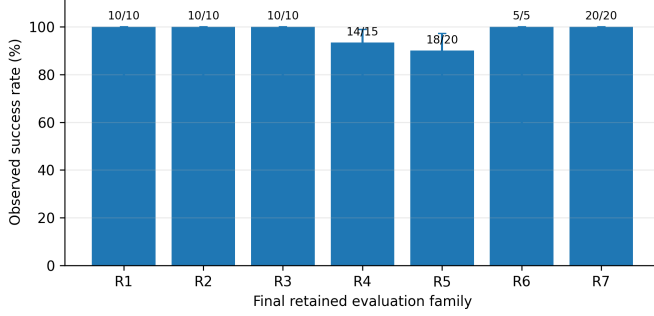


Fig. 2. Family-wise success rates with 95% Wilson intervals. Percentages based on small episode counts should not be interpreted as proof of a 100% underlying success probability.

are horizon/configuration sensitive. We therefore retain the authoritative 5 s master result.

B. Controlled component ablation

To isolate component contributions without conflating them with the 90-episode regression, five controller variants were evaluated on the same fixed 14-episode representative subset (Table IV). The identical episodes and seeds were held fixed across all variants and the classical baseline, providing scenario-matched comparisons intended to isolate controller-component effects rather than replace the broader validation suite. The nominal recurrent PPO alone completed 9/14 episodes and collided five times. Adding the residual policy without the safety filter did not provide safety supervision and reduced success to 6/14 with eight collisions. In contrast, nominal PPO plus the predictive safety filter completed 14/14 episodes without collision, identifying predictive screening as the principal safety mechanism on this subset. The complete architecture without progress-aware ranking completed 13/14 episodes with one collision. Adding progress-aware ranking produced 14/14 collision-free completion, increased mean speed from 1.187 to 1.281 m/s, and reduced measured low-speed dwell (time below 0.2 m/s) from 1.58 s to 0.00 s. These are fixed-subset observations rather than claims of universal statistical superiority. Wilson 95% intervals further reflect the limited subset size: 9/14 gives 38.8–83.7%, 6/14 gives 21.4–67.4%, 13/14 gives 68.5–98.7%, and 14/14 gives 78.5–100%.

C. Classical baseline comparison

A classical non-learning baseline combining 4 m look-ahead line-of-sight (LOS) guidance with heading/yaw-rate feedback and reactive CPA/TCPA collision avoidance was evaluated

TABLE IV
COMPONENT ABLATION ON THE FIXED 14-EPISEDE SUBSET

Variant	Succ.	Coll.	Min clr. (m)	CTE (m)	Low-speed	
					$\bar{U}$ (m/s)	dwell (s)
A Nominal PPO	9/14	5	-0.151	0.697	0.925	11.19
B + residual	6/14	8	-0.193	0.699	0.953	0.00
C + safety filter	14/14	0	0.285	0.806	1.193	8.99
D Full, no progress	13/14	1	-0.059	0.781	1.187	1.58
E Proposed full	14/14	0	0.273	0.955	1.281	0.00

TABLE V
CLASSICAL COMPARISON ON THE SAME FIXED 14-EPISEDE SUBSET.

Method	Succ.	Coll.	Path		CTE (m)	$\bar{U}$ (m/s)
			limit	Min clr. (m)		
LOS+CPA/TCPA	10/14	2	2	-0.022	2.679	1.182
Nominal PPO	9/14	5	0	-0.151	0.697	0.925
Proposed full	14/14	0	0	0.273	0.955	1.281

on the identical 14-episode subset (Table V). It achieved 10/14 successes (71.4%; Wilson 95% CI: 45.4–88.3%). Its four unsuccessful episodes comprised two collisions and two cross-track-limit terminations, where the common environment terminated the episode at $|CTE| > 20$ m under collision risk < 0.2 . The nominal recurrent PPO achieved 9/14 successes (64.3%; Wilson 95% CI: 38.8–83.7%) with five collisions despite the lowest mean $|CTE|$ (0.697 m). The proposed full method completed 14/14 episodes (100%; Wilson 95% CI: 78.5–100%) without collision, with mean $|CTE| = 0.955$ m and mean speed 1.281 m/s. Thus, on this fixed subset, the proposed architecture improved completion and collision avoidance relative to both the conventional reactive baseline and the unprotected learned controller while retaining substantially lower tracking error than the classical baseline.

D. Post-training generalization and progress ranking

All six documented representative post-training scenarios completed successfully. Representative outcomes include mean $|CTE| = 0.228$ m for the clean circle, minimum static clearance 0.495 m for the four-obstacle circle, minimum dynamic clearance 0.447 m for the five-vessel shoreline case, and minimum static clearance 2.653 m through the synthetic archipelago. The mixed rocks/vessels scenario reached approximately the 0.20 m static margin without collision.

The most informative case is the circular mixed-hazard scenario in Fig. 3. Before the progress term in Eq. (11), the safety filter could repeatedly prefer very low-speed safe actions and fail to make useful progress. With the progress term added, the same high-level controller family completed the scenario in 1734 steps with minimum dynamic clearance approximately 0.958 m, minimum static clearance approximately 0.20 m, and no collision or timeout. The safety margin, prediction horizon, learned policies, candidate set, and emergency logic were unchanged; only ranking among already-safe actions changed.

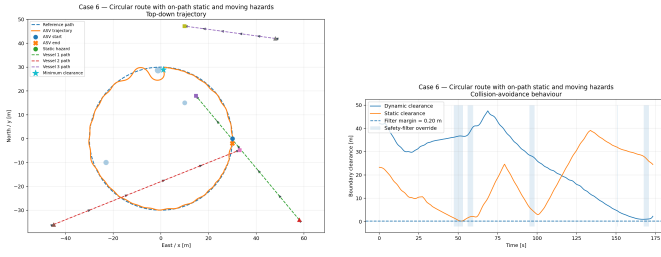


Fig. 3. Representative post-training mixed-hazard Case 6 after progress-aware ranking. Left: trajectory with static and moving hazards. Right: dynamic/static clearance and safety-filter intervention.

TABLE VI
POST-TRAINING GENERALIZATION SCENARIOS

Case	Scenario	Outcome
1	Clean 30 m circular path	Success; 1275 steps; mean $ CTE $ 0.228 m
2	Circle + four static hazards	Success; min. static clearance 0.495 m
3	Shoreline + five moving vessels	Success; min. dynamic clearance 0.447 m
4	Three synthetic islands	Success; min. static clearance 2.653 m
5	Three rocks + two vessels	Success; min. static clearance 0.200 m
6	Circle + mixed static/dynamic hazards	Success; 1734 steps; min. dynamic clearance 0.958 m; min. static clearance 0.200 m

E. Computational performance

A CPU profiling benchmark on the three-vessel stress family (R3) measured 1285 controller steps after discarding the first 20. Mean inference times were 0.824 ms (speed PPO), 1.246 ms (steering PPO), and 0.658 ms (residual PPO), for about 2.73 ms total learned-policy inference. The safety filter dominated runtime: median filter latency was 5.06 ms but mean latency was 38.44 ms, P95 was 516.45 ms, and maximum was 666.82 ms. End-to-end latency had median 7.57 ms, mean 41.23 ms, P95 519.26 ms, and maximum 669.70 ms; 5.99% of steps exceeded the nominal 100 ms control period. The episode nevertheless completed successfully. Thus learned-policy inference is lightweight, while predictive-search tail latency is the main implementation target before deterministic real-time hardware execution.

IX. DISCUSSION AND LIMITATIONS

The controlled ablation strengthens the architectural interpretation of the full regression. On the fixed 14-episode subset, predictive safety filtering was the principal observed safety mechanism: nominal PPO changed from 9/14 with five collisions to 14/14 without collision when protected by the filter. Residual correction alone was not a safety mechanism. Within the complete architecture, progress-aware ranking changed 13/14 with one collision to 14/14 collision-free completion, increased mean speed, and eliminated measured low-speed dwell on this subset. The same episodes and seeds were deliberately held fixed across every ablation arm and the classical controller so that differences are scenario-matched; because the subset is small, these results establish component behavior under controlled fixed scenarios rather than population-level superiority. The tracking result also exposes

TABLE VII
CPU TIMING OF FINAL CONTROLLER (R3, 1285 MEASURED STEPS)

Component	Mean	Median	P95	Max
Speed PPO	0.824	0.754	1.246	27.764
Steering PPO	1.246	1.157	1.939	6.693
Residual PPO	0.658	0.581	0.976	4.208
Safety filter	38.437	5.055	516.445	666.815
End-to-end	41.225	7.565	519.260	669.699

Times in milliseconds; 5.99% end-to-end steps exceed 100 ms.

the intended trade-off: nominal PPO attained lower mean $|CTE|$ when it remained viable, whereas the protected full architecture accepted modestly larger path deviation while achieving collision-free completion and higher mean speed on this subset.

The present paper reports the simulation-validation stage of an autonomy controller developed from target-ASV operating characteristics and transferred for subsequent hardware validation. The reduced-order planar/Nomoto model deliberately isolates guidance, residual coordination, and predictive collision avoidance; HIL and controlled water trials are the next validation stage rather than evidence claimed in this paper. The COLREG component is limited to encounter features and a soft starboard preference, so the results should be interpreted as COLREG-aware maneuver evaluation rather than formal regulatory certification. The robustness families also identify remaining engineering work: tracking robustness reached 14/15 and the environmental-disturbance sweep 18/20, motivating stronger maneuver-reachability and disturbance-robust prediction. CPU profiling further shows that candidate-search tail latency should be reduced through pruning, parallel rollout evaluation, or bounded-time search before deterministic onboard execution. The classical comparison is intentionally limited to one conventional non-learning controller on the same 14 fixed episodes; broader published-method benchmarking and larger matched-seed evaluations remain useful follow-up work.

X. CONCLUSION

A hierarchical ASV autonomy architecture combining recurrent nominal steering and speed policies, recurrent joint residual correction, and predictive steering-speed safety filtering was evaluated in mixed static and multi-vessel scenarios. A progress-aware objective applied only to already-safe candidates reduced stop-heavy behavior in a representative post-training mixed-hazard case while leaving the explicit safety test unchanged. The seven-family simulation validation achieved 87/90 successes (96.67%) with no stalls, and the documented post-training generalization scenarios completed successfully. The controller and evaluation package were transferred as the simulation-validated autonomy baseline for subsequent HIL integration and controlled water-trial validation. A controlled 14-episode ablation identified predictive filtering as the primary safety mechanism on that subset and showed an observed benefit from progress-aware ranking; the proposed method also completed 14/14 episodes versus 10/14 for a classical LOS+CPA/TCPA baseline. Remaining technical work includes

larger-scale statistical evaluation, broader external benchmarking, robustness to tracking and environmental disturbance, deterministic runtime optimization, and actuator-aware hardware validation.

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